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# An integrated Controller-Estimator technique for Vector control and Speed estimation in PMSM Drive

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## ABSTRACT

This study presents an integrated controller and estimator technique, using a stationary reference frame ( $\alpha\beta$ ) system, for a surface permanent magnet synchronous motor (PMSM) drive system. This paper aims to develop an improved sensorless vector control where the Field Oriented Control (FOC) method and the speed estimation are performed together, as opposed to being separate functions in the conventional method for a PMSM drive. In the proposed technique, current control and motor speed - rotor position estimations are performed based on current sensor feedback only. The speed is estimated from  $\alpha\beta$ -currents of the motor, thereby eliminating the need for speed sensor. The current control loop of the vector control is implemented using  $\alpha\beta$ -currents instead of dq-currents. The main advantages of this proposed approach are reduced circuitry and simple control configuration. The key benefits include reduction of sensors, controller complexity, computing complexity, and cost. The proposed integrated approach is theoretically investigated and simulated for the PMSM drive model. Using the Opal-RT configuration, the performance is also assessed in real-time. The proposed control strategy shows promise in meeting the closed-loop drive system's speed tracking requirements. A comparison of the simulation results between the conventional and proposed models validates the effectiveness of the proposed technique for speed control applications. When compared to the conventional control method, the proposed model's performance improved in terms of different transient levels, speed responses and evaluation metrics.

## KEYWORDS

Current controller, Permanent magnet synchronous motor (PMSM), Vector control, Speed estimation,  $\alpha\beta$ -axes.

## I. INTRODUCTION

Vector control is the commonly used method for speed control of Alternating Current (AC) motors because of better motor response over a wide speed range and high starting torque. Among different vector control techniques, the Field Oriented Control (FOC) is considered superior due to high motor performance at low speeds. An indirect FOC is more commonly used for motor control. In this method, the feedback current signals are converted into d-axis and q-axis current commands and used to generate voltage commands [1-4]. These voltage commands are used to generate pulse signals to the inverter module which supplies three-phase voltages to the permanent magnet synchronous motor (PMSM) [5,6].

An overview of the existing FOC techniques and speed estimation methods for AC motors is given in [1-4]. The architecture of a digital controller module is explained in [5]. It also uses SMO (sliding mode observer) for speed estimation, which requires both voltage and current feedback. A unified FOC controller is introduced in [6], which uses dq-axis PI controllers for current control. Machine Learning (ML)-based controllers with algorithms like Linear Regression, Support Vector Machine regression, Artificial Neural Network (ANN), and deep learning technique such as Long Short-Term Memory network are presented in [7], which use dq-axis system for current control. A sensorless FOC technique for PMSM with angle-tracking Phase-Locked Loop (PLL) estimator is discussed in [8], while SMO technique for speed estimation is discussed in [9], and BEMF (Back Electromotive Force) observer and tracking observer are used in [10] for sensorless operation. To estimate speed, a combination of ANN and PLL methods is used in [11] to derive the relations between rotor angle with voltage and current feedback signals in the  $\alpha\beta$ -axis system. In [12], a single current sensor and Model Reference Adaptive Standard (MRAS) method are used for speed estimation, and dq-axis PI controllers for current control. In [13], feedback currents in dq-axis system are used for PI-based current control and, for speed and position estimation with an adjustable model and adaptation mechanism. The BEMF method and MRAS method are used in [14] for speed estimation, and the dq-axis system is used for current controllers. The main limitation of all these methods is the requirement for separate subsystems for current control and speed estimation.

Sensorless speed estimation techniques have the advantages of reduced circuitry, improved system reliability, and reduced cost. They are broadly classified as Signal Injection (SI) methods and Model-based methods. SI methods are more effective in zero and low speed ranges, and easy to implement. However, the signal injections cause voltage and current distortions. Model-based methods are applied at medium and high speed

ranges, and are divided as open loop estimators and closed loop estimators [1-4]. Open loop estimators do not require any tuning but they suffer from poor robustness and high sensitivity. Popular techniques of closed loop estimators include filters, sliding mode observers, model reference adaptive systems, and artificial intelligence methods [5,8-14]. SMO is a robust technique with fast transient response; but it has drawbacks such as chattering, and complex design of SMO gain. MRAS is a simple technique with wide operating range and supports multi-parameter estimation. However, the gain tuning process is very time-consuming. Filters have the advantage of anti-noise capability, but increases the computational load. AI methods depend less on the model, though they are limited by space constraints and increased computation [15,16].

In the field of vector control and speed estimation of PMSM, the methodologies proposed by existing literature require separation of the current control and estimation functions. The current control requires dq-axis current signals or voltage vectors. Some methods use voltage and current sensors for speed estimation while few methods use complicated techniques like MRAS, PLL, and SMO for the same. The existing literature also highlighted the absence of a more compact system.

The methodology proposed in this paper minimizes the circuit connections and integrates the two separate functions of current control and estimation into a single subsystem. The proposed technique uses the  $\alpha\beta$ -axis system, thereby using the same set of inputs for the current control and estimation. This eliminates the need for separate input sets for the two functions, resulting in reduced computation. It does not require any complex techniques to estimate rotor position and speed, as the relationship between  $\alpha\beta$ -axes and trigonometric functions is directly used to derive the rotor position and speed.

The following novelties are introduced by this paper:

- 1) The proposed technique performs the dual function of current control and speed estimation, instead of performing them separately, thereby reducing the computational complexity of the controller.
- 2) The proposed technique gives a cost-effective and compact system, as the number of sensors and subsystems required are reduced.
- 3) The estimation process proposed is simpler, compared to existing techniques.
- 4) This work also introduces a machine learning-based model for the proposed integrated technique, thereby exploring the field of Artificial Intelligence.

The study is organized as follows: Section II and Section III describe the standard vector control mechanism, the proposed method, and theoretical details about PMSM drives. Section IV displays the performance analysis of the proposed models under several criteria, Section V discusses the findings, and Section VI presents the conclusion.

## II. CONVENTIONAL METHODOLOGY

### A. PMSM MODEL

In the synchronously rotating coordinate system (dq-axes), the PMSM's stator voltage equations [7,17] are provided by

$$\begin{pmatrix} v_{sd} \\ v_{sq} \end{pmatrix} = \begin{pmatrix} R_s + L_d \frac{d}{dt} & -\omega_e L_q \\ \omega_e L_d & R_s + L_q \frac{d}{dt} \end{pmatrix} \begin{pmatrix} i_d \\ i_q \end{pmatrix} + \begin{pmatrix} 0 \\ \omega_e \psi_f \end{pmatrix} \quad (1)$$

The electrical torque produced in the PMSM drive is represented by

$$T_e = J_{eq} \frac{d\omega_m}{dt} + B_a \omega_m + T_L \quad (2)$$

$$T_e = \frac{3P}{2} [\psi_f i_q + (L_d - L_q) i_d i_q] \quad (3)$$

where,

$v_{sd}, v_{sq}$  = instantaneous values in dq-axes for stator voltages

$i_{dq}$  = instantaneous values in dq-axes for stator currents

$R_s$  = stator winding resistance

$L_{dq}$  = stator winding inductance values in dq-axes

$\omega_e$  = motor speed in electrical terms

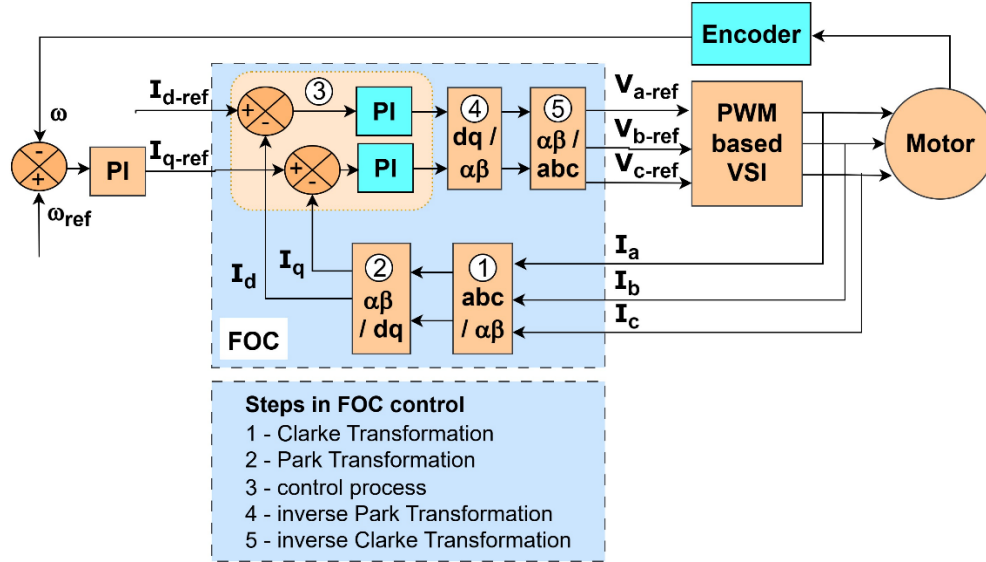
$\omega_m$  = motor speed in mechanical terms

$\psi_f$  = flux linkage of the PM rotor

$J_{eq}$  = inertia of the motor

$B_a$  = the frictional coefficient of motor

$T_L, T_e$  = load torque and electromagnetic torque of the motor drive



**FIGURE 1. Vector control : Conventional approach**

$P$  = number of pole pairs.

In case of a surface PMSM motor,  $L_d = L_q$ . Hence the second term becomes zero in (3). The relation between electrical rotor angle  $\theta_e$ , and motor speeds  $\omega_m$  and  $\omega_e$  is given by

$$\omega_e = \omega_m * P \quad (4)$$

$$\theta_e = \int \omega_e dt \quad (5)$$

Symmetric three-phase systems, sinusoidal back-emf, hysteresis loss and eddy current loss neglect, constant coil resistance and reactance, and insignificant harmonics are some of the assumptions used in PMSM drive modeling.

### **B. FOC CONTROL OF PMSM DRIVE**

The field-oriented control (FOC) or the vector control is a simple technique that treats a three-phase PMSM drive system as a separately excited DC motor, by separating stator currents into two orthogonal d-q axis currents. The PMSM drive's FOC schematic and the steps in FOC technique are shown in Fig. 1.

The stator currents of the motor in the 3-phase co-ordinate system (abc) are transformed into 2-axis synchronous rotating co-ordinate system (dq) through the combination of Clarke Transformation (abc to  $\alpha\beta$ ) and Park Transformation ( $\alpha\beta$  to dq), i.e., steps 1 and 2. By this process, the stator current is divided into torque component current (along q-axis) and flux component current (along d-axis). In step 3, a control process is performed on the dq-axis currents, usually with PI controllers. This results in producing the dq-axis reference voltage signals. Through steps 4 and 5, these signals are transformed into 3-phase voltage signals which are then used for inverter pulse generation.

The FOC consists of two cascaded control loops: the outer loop for speed control, and two inner control loops for current regulation. The transfer function model for developing the controller in a speed-loop can be constructed by rearranging (2) and (3).

$$TF_{\omega} = \frac{\psi_f \cdot P}{(B_a + s \cdot J_{eq})} \quad (6)$$

Rearranging (1) provides the governing equations to construct the PI controllers in the current loop. In this stage, the compensation terms  $(-\omega_e L_q i_q, \omega_e L_d i_d, \omega_e \Psi_f)$  are disregarded. The compensation terms are used to decouple the stator current into components of torque and flux current during the control process. They are subsequently added to the d-axis and q-axis PI controllers' outputs to create the final current-loop control setup. This avoids the inaccurate results caused by decoupling [2,7,18-20]. The conventional approach uses the Ziegler-Nicholas methodology to set the gains of the PI controllers [21].

$$TF_{id} = \frac{1}{(R_s + s \cdot L_d)} \quad (7)$$

$$TF_{iq} = \frac{1}{(R_s + s \cdot L_q)} \quad (8)$$

### III. PROPOSED METHODOLOGY

#### A. PROPOSED APPROACH

The proposed approach addresses two separate functions of the motor drive – current control and speed estimation – and combines them into a single function, that uses only current sensors. In this technique, only the current feedback is used for current regulation as well as speed estimation. This eliminates the need for speed sensors and voltage sensors, thereby reducing the circuit components and increasing the compactness of the overall motor drive [22]. Fig.2 gives the overall system architecture of the proposed technique.

Fig.3 shows the operation of the integrated Controller-Estimator (iCE). The dq-axis reference currents generated from speed PI controller are converted to  $\alpha\beta$ -axis reference currents by Park's inverse transformation and given to the iCE unit (inputs 1 & 2). The 3-phase feedback current signals are converted to  $\alpha\beta$ -axis currents by Clarke transformation, and given as next set of inputs to iCE unit (inputs 3 & 4). The feedback currents are used for both controller and estimator functions, while the reference currents are required only for the controller function.

The controller function performs comparison of the feedback signals and reference signals, and the errors in  $\alpha\beta$ -axis currents are given as inputs  $e_a$  and

$e_b$ , as seen in (9) and (10), to two PI controllers. The PI controllers in the current loop are tuned in the same way as that of the conventional method. The output voltages of the PI controllers are converted to dq-axis values, which are then added with decoupling terms and thus forms the first set of outputs, i.e. voltage reference commands for pulse generation.

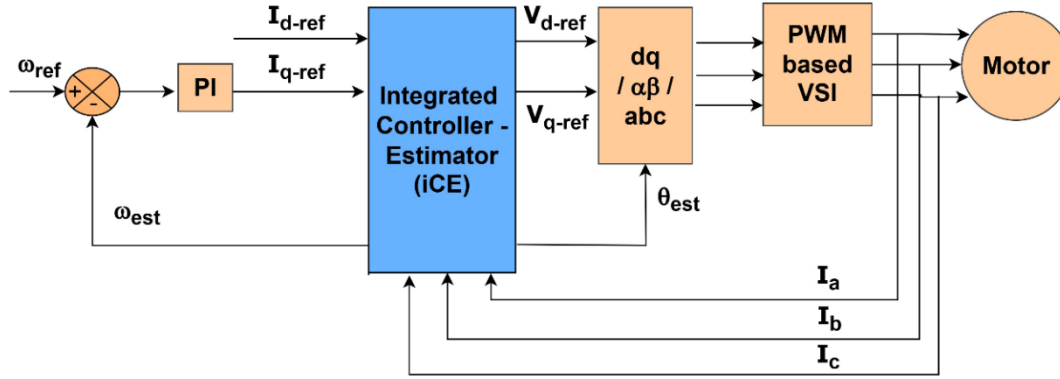


FIGURE 2. Proposed control technique - using PI controllers

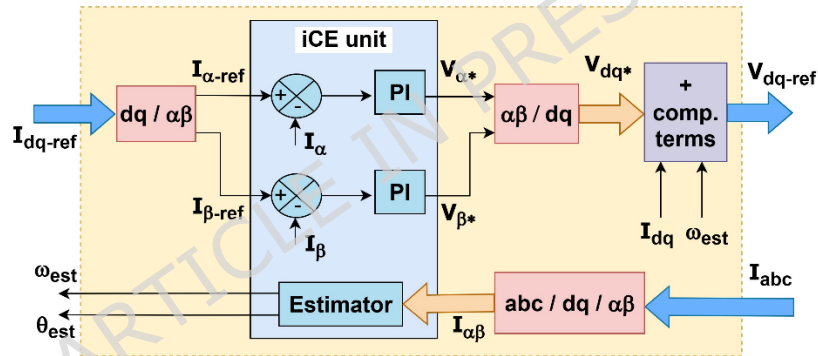


FIGURE 3. Integrated Controller-Estimator (iCE)

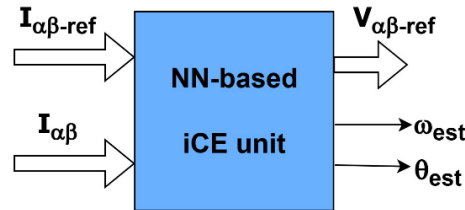
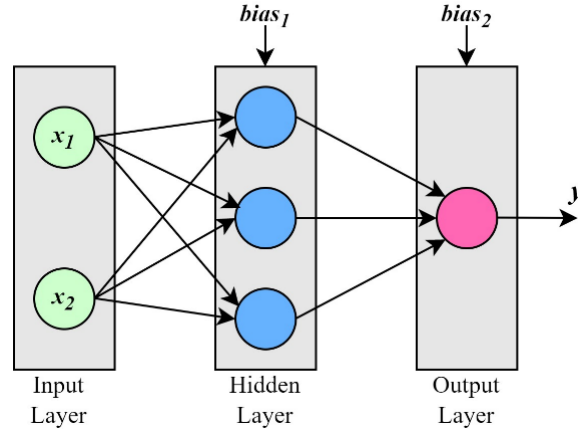
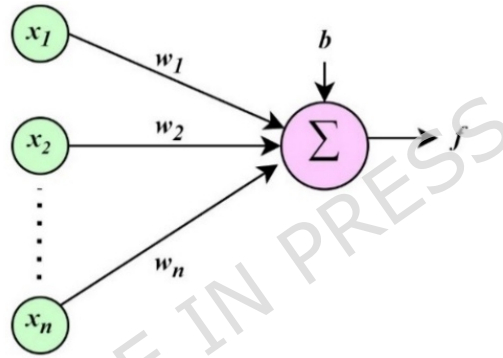


FIGURE 4. Proposed control technique - using NN



**FIGURE 5. NN architecture**



**FIGURE 6. Layout of an artificial neuron**

$$e_{\alpha} = I_{\alpha\text{-ref}} - I_{\alpha} \quad (9)$$

$$e_{\beta} = I_{\beta\text{-ref}} - I_{\beta} \quad (10)$$

The estimator function uses the relation between  $\alpha\beta$ -axes and trigonometric functions to estimate the speed and rotor position. For a symmetric 3-phase system, if the feedback currents are given by (11), then the transformed currents in  $\alpha\beta$ -axes are given by (12). By rearranging (12) and applying inverse trigonometric functions, rotor angle can be estimated from the feedback currents and this value is used in (13) to estimate speed. The remaining outputs of the iCE unit are thus obtained.

$$\begin{pmatrix} I_a \\ I_b \\ I_c \end{pmatrix} = I_{\text{peak}} \begin{pmatrix} \cos \theta \\ \cos(\theta - 120^\circ) \\ \cos(\theta - 240^\circ) \end{pmatrix} \quad (11)$$

$$\begin{pmatrix} I_{\alpha} \\ I_{\beta} \end{pmatrix} = I_{\text{peak}} \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} \quad (12)$$

$$\omega_e = \frac{d\theta_e}{dt} \quad (13)$$

### B. ML-BASED MODEL OF PROPOSED APPROACH

Fig.4 shows a second proposed model, which substitutes a machine learning (ML)-based iCE unit for the PI controller-estimator configuration in the first proposed model. The purpose of this model is to explore the possibility of ML-based versions of the proposed iCE model. As an initial step, a feedforward neural network (NN)-based iCE unit is studied. The NN model is trained and tested using data from the model of Fig.2. Here the NN-based unit takes the current signals, and generate the voltage commands in  $\alpha\beta$ -axis system and trigonometric functions of rotor angle as outputs. The voltage commands are converted to dq-axis system for pulse generation, while rotor angle and speed are estimated from the other outputs.

The NN-based iCE model is trained offline with dataset that include constant conditions and varying conditions of speed and load. For testing parameter variations, the same model was used without retraining. The training data of 800,000 samples has a sampling time of 1  $\mu$ sec; 75% of this is used for training the model and 25% is used for validating the model. The testing data of 10,000 samples has a fixed sampling time of 50  $\mu$ sec.

Information can only move from the input to the output in a feedforward neural network (NN). Three levels—input, output, and hidden layers—are used to structure the artificial neurons in neural networks, which are based on biological brain models. The depth of the network is determined by the number of hidden layers. A weight is assigned to each neural connection, and each layer has a bias [7,23].

A NN model with two inputs, one output, and a single hidden layer with three neurons is depicted in Fig. 5. In this case, the inputs are  $x_1$  and  $x_2$ , and the output is  $y$ . An artificial neuron's output, as shown in Fig. 6, is a function  $f$  of the sum of any weights ( $w_1$  to  $w_n$ ) from inputs ( $x_1$  to  $x_n$ ) and bias  $b$  [7,24]. Normalizing a neuron's output is accomplished by using the activation function  $f$ , which is provided by (14).

$$y = f(b + \sum_{i=1}^n x_i w_i) \quad (14)$$

### C. DEVELOPMENT OF NN-BASED MODEL

MATLAB's *train* function is utilized to train a basic neural network with the following parameters set:

- *Number of hidden layers* and *Size of each hidden layer*,
- *Activation function* of each layer,
- *Training algorithm*.

Starting with the lowest values, the *number* and *size* of the hidden layers are chosen through a trial-and-error process. With six neurons in the first layer

and four neurons in each of the other layers, a good outcome was achieved at three hidden layers.

The *activation functions* for the output layer and hidden layer are the linear function and the hyperbolic tangent sigmoid transfer function, respectively.

Neural network *training algorithms* in MATLAB include the Levenberg-Marquardt, Bayesian Regulation, Gradient descent, and Resilient Back Propagation (RPROP) algorithms, with the Levenberg-Marquardt algorithm being the most popular and fastest.

#### D. EVALUATION METRICS

The performance of the proposed models is evaluated using three metrics: mean absolute error (MAE), root mean square error (RMSE), and symmetric mean absolute percentage error (SMAPE). These are common statistical metrics used to assess how well time-series-based models work [7,24-26].

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |(\bar{y}_i - y_i)| \quad (15)$$

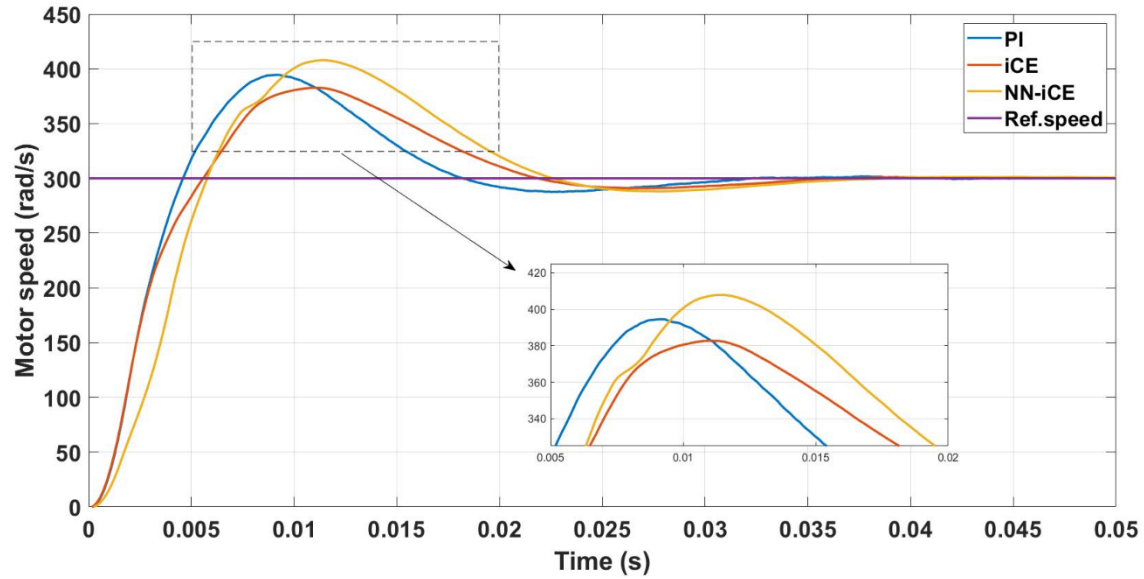
$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\bar{y}_i - y_i)^2} \quad (16)$$

$$\text{SMAPE} = \frac{100\%}{N} \sum_{i=1}^N \frac{|(\bar{y}_i - y_i)|}{\left( \frac{|\bar{y}_i| + |y_i|}{2} \right)} \quad (17)$$

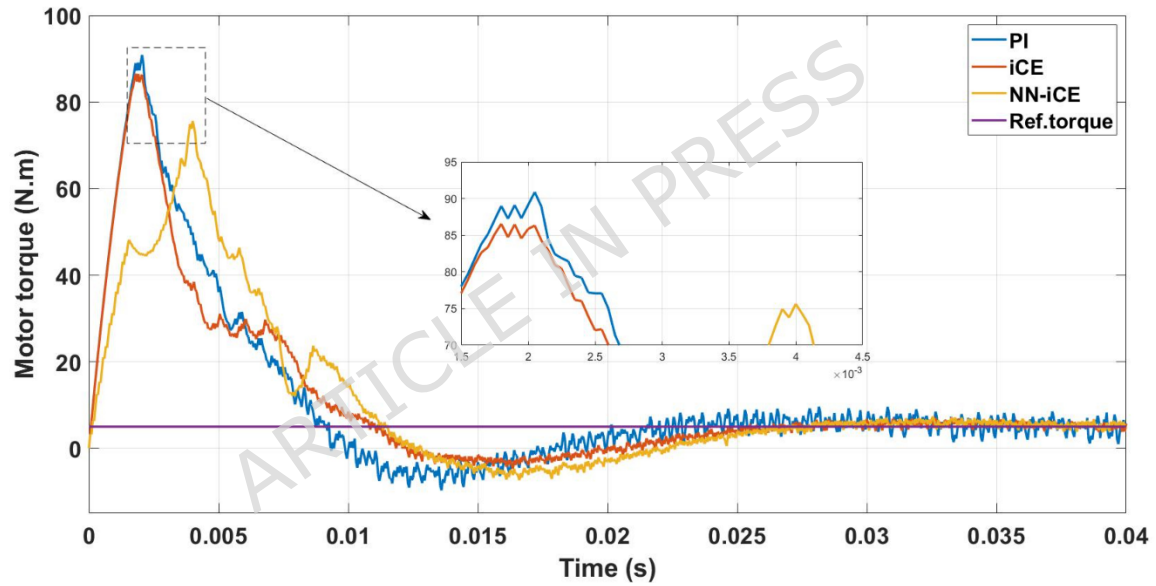
Here  $y_i$  is the real value,  $\bar{y}_i$  is the predicted value and  $N$  is the sample size.

TABLE 1  
MOTOR PARAMETERS

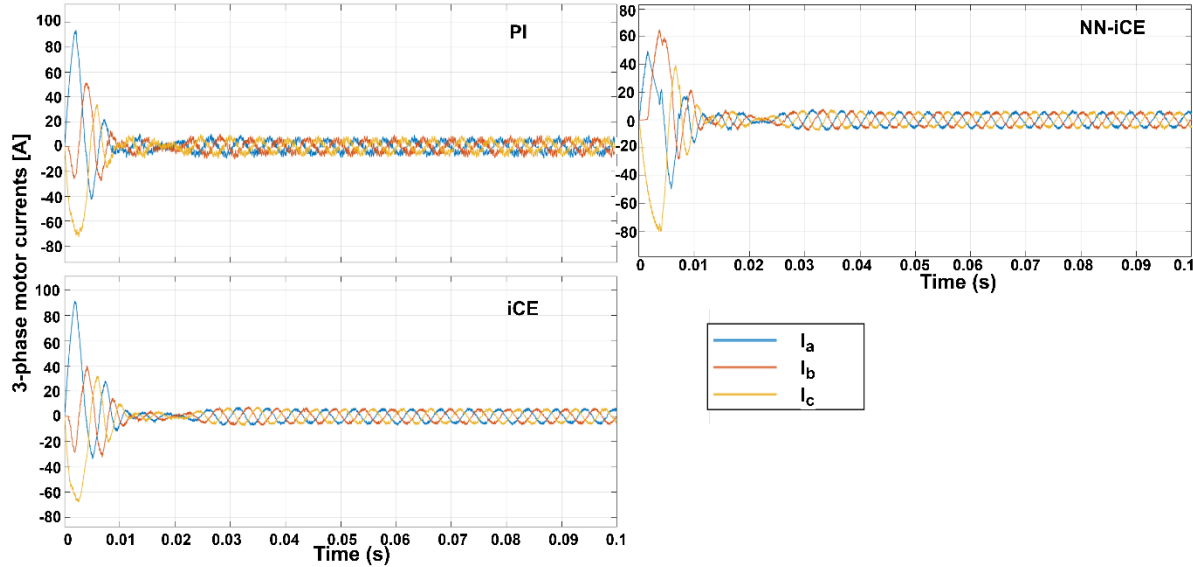
| PARAMETER                       | VALUE                     |
|---------------------------------|---------------------------|
| $R$ (Resistance)                | 2.85 $\Omega$             |
| $L_{dq}$ (Stator inductances)   | 0.005 H                   |
| $B_a$ (Coefficient of friction) | 1.0 e-4 Nm-s              |
| $J_{eq}$ (Motor inertia)        | 8.0 e-4 kg-m <sup>2</sup> |
| $P$ (Pole pairs)                | 4                         |
| $\Psi_f$ (Rotor flux linkage)   | 0.1548 Wb-turms           |
| $\omega_m$ (Rated speed)        | 300 rad/s                 |



**FIGURE 7. Speed response at rated conditions: Conventional vs Proposed models**



**FIGURE 8. Torque response at rated conditions: Conventional vs Proposed models**



**FIGURE 9. Current response at rated conditions: Conventional vs Proposed models**

#### IV. EVALUATION OF PROPOSED TECHNIQUES

A MATLAB/Simulink environment is used to create the training and testing data from the model of the conventional system. MATLAB Simulink and Coder are used to compare the performance of the proposed and conventional approaches. The PMSM parameters are listed in Table 1. Subsections A through E provide specifics on the testing of the proposed models [7,21].

##### A. SIMULATION RESULTS

The conventional model of FOC technique which uses PI controllers is shown in Fig. 1. For comparison purposes, the same PI values for speed and current controllers are used for the conventional and proposed models. Two models are proposed here, where iCE is the main proposed model and NN-based iCE is the ML variant, or a modified version, of this proposed model.

##### 1) CONSTANT SPEED AND CONSTANT LOAD

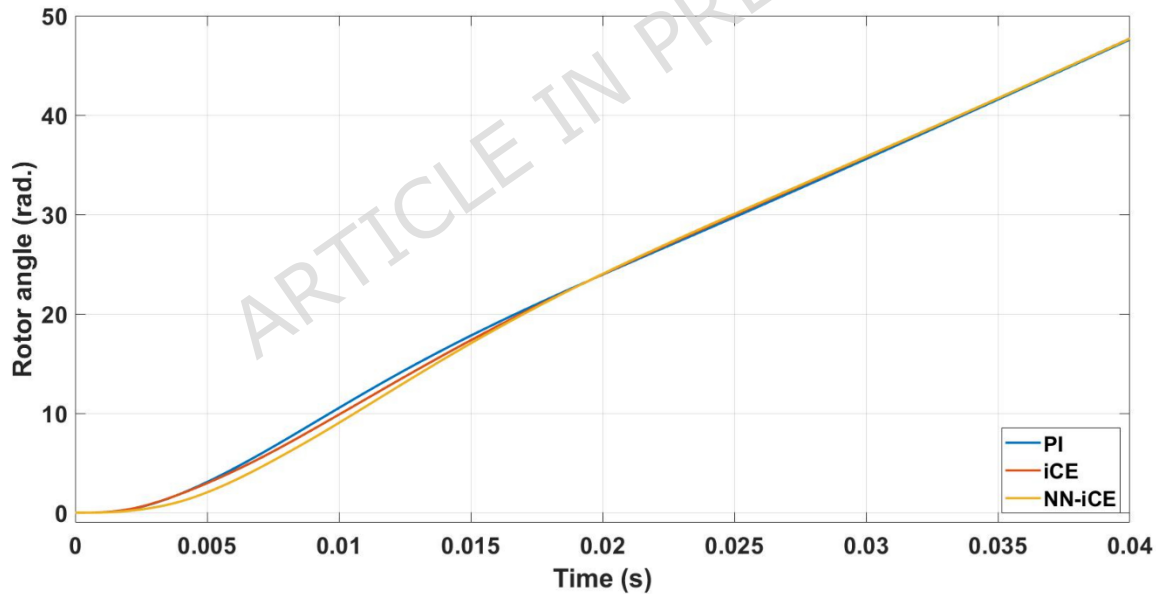
Fig. 7 gives the speed responses of the conventional model and the two proposed models (iCE and NN-iCE) for a fixed speed of 300 rad/s and load of 5 Nm. Fig. 8 gives the torque response comparison of the models. Fig. 9 shows a comparison of the 3-phase motor currents while Fig. 10 gives a comparison of the rotor angles determined from the estimated speed values. Table 2 gives the comparison results of the speed, torque and current responses under rated conditions. Least values are highlighted in bold.

The iCE model gives the least overshoot for speed (1.275%) and NN-based iCE model has the least transient peak for torque (75.64 Nm). However conventional model has faster settling time for both speed and torque responses (0.032s and 0.022s respectively). In the case of current response, though all the models settle at the same time, the proposed models draw lesser

current (6A) than the conventional model. The rotor angle values are similar for all the models. The angles are same after reaching steady state, while the angles during transient state estimated by the proposed models lag slightly.

TABLE 2  
COMPARISON OF RESPONSES UNDER RATED CONDITIONS

| COMPARISON                | DESCRIPTION          | PI           | iCE          | NN-iCE       |
|---------------------------|----------------------|--------------|--------------|--------------|
| Speed response (Fig. 7)   | Peak overshoot (%)   | 1.315        | <b>1.275</b> | 1.359        |
|                           | Settling time (s)    | <b>0.032</b> | 0.036        | 0.038        |
| Torque response (Fig. 8)  | Peak overshoot (Nm)  | 91           | 86.5         | <b>75.64</b> |
|                           | Settling time (s)    | <b>0.022</b> | 0.025        | 0.026        |
| Current response (Fig. 9) | Settling current (A) | 8            | <b>6</b>     | <b>6</b>     |
|                           | Settling time (s)    | 0.04         | 0.04         | 0.04         |



**FIGURE 10. Rotor angle at rated conditions: Conventional vs Proposed models**

## 2) SPEED VARIATIONS

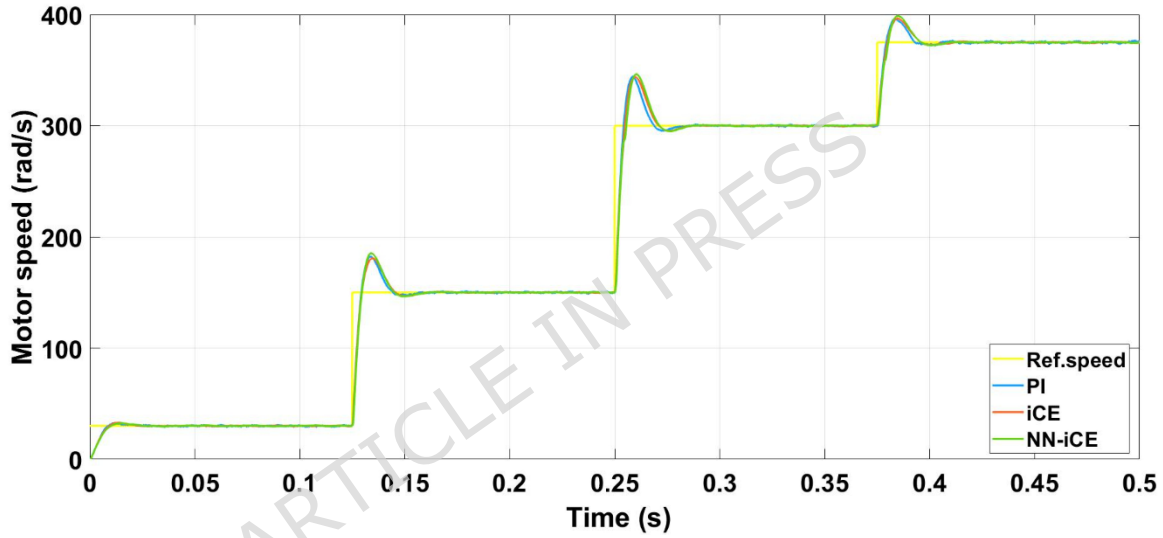
The conventional and proposed models are subjected to increasing speed values of 10%, 50%, 100% and 125% of the rated speed, at equal intervals of 0.125 sec. The speed response in this case is given by Fig. 11. The responses of all the models are similar in this case.

## 3) LOAD VARIATIONS

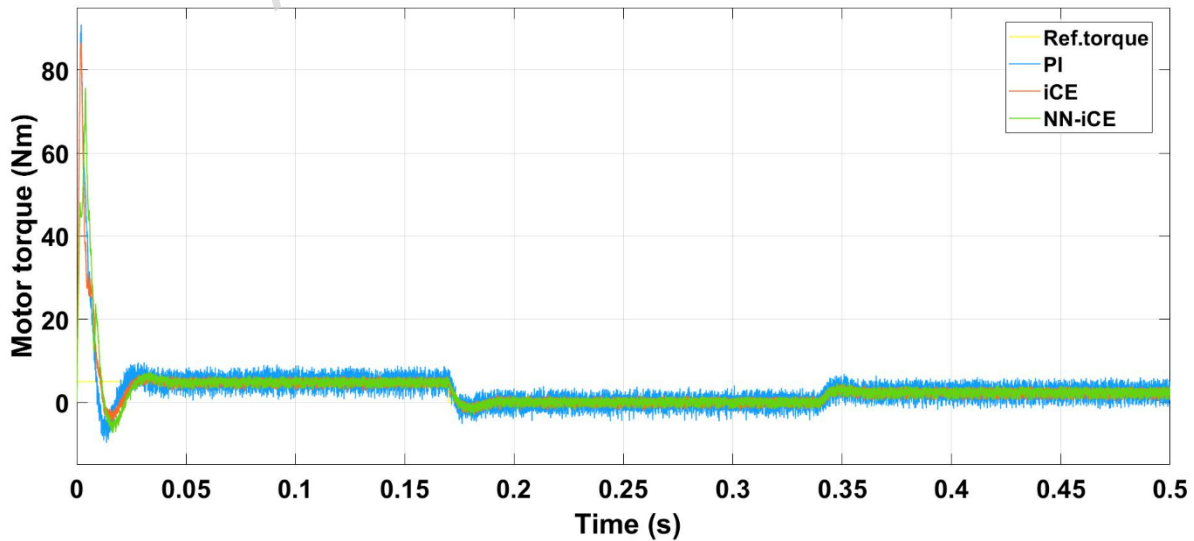
The load applied to the conventional and proposed models are varied from full-load to no-load and then to half-load, at equal intervals of 0.17 sec. Fig. 12 and Fig. 13 give the torque and speed responses in this case, respectively. All the models respond similarly in this case.

#### 4) MOTOR PARAMETER VARIATIONS

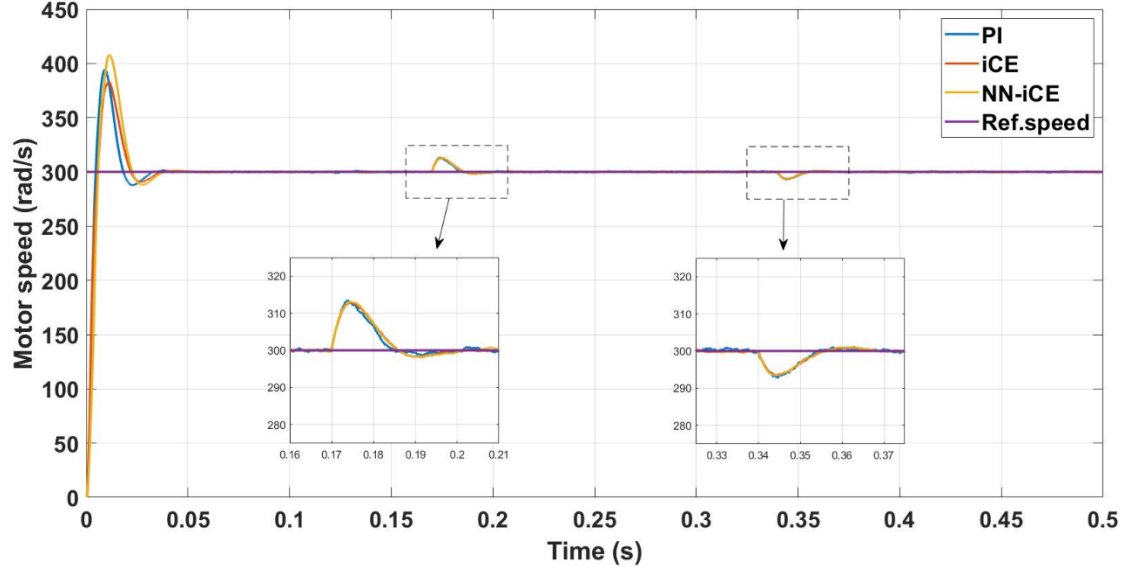
The robustness of the proposed models in the FOC method are tested by varying (i) mechanical parameters  $J_{eq}$ ,  $B_a$  (ii) electrical parameters  $R$ ,  $L_{dq}$  and (iii) magnetic flux linkage  $\Psi_f$  separately. Fig. 14 - Fig. 16 give the speed responses for each case, and it is seen that the proposed models withstand the parameter variations, similar to the conventional model. Table 3 gives a comparison of settling time and transient peaks for each case, between the conventional and proposed models.



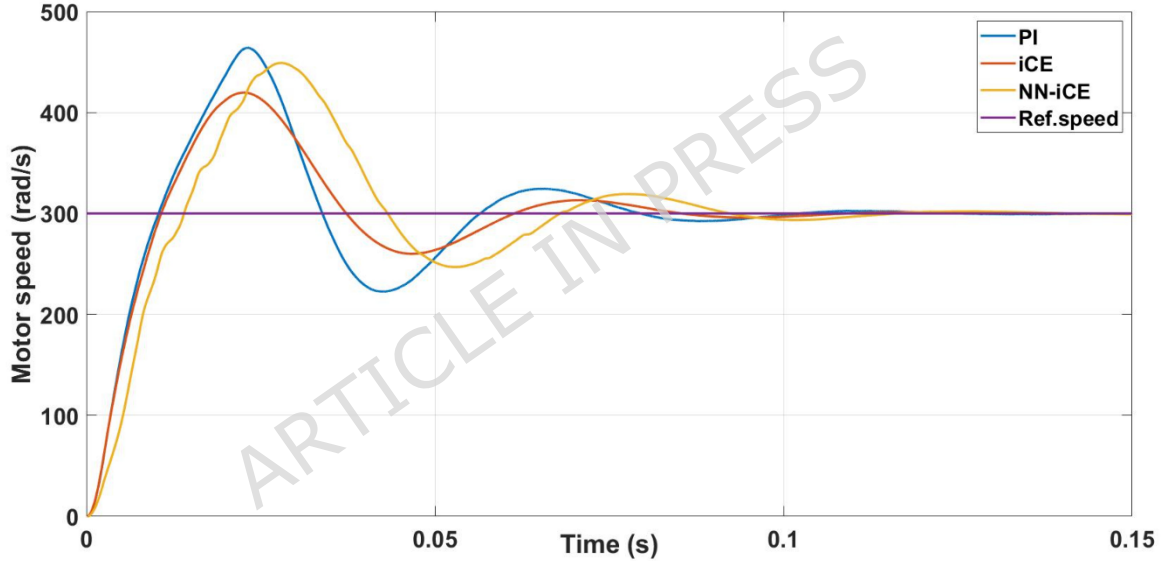
**FIGURE 11. Speed response for varying speeds: Conventional vs Proposed models**



**FIGURE 12. Torque response for varying loads: Conventional vs Proposed models**



**FIGURE 13. Speed response for varying loads: Conventional vs Proposed models**



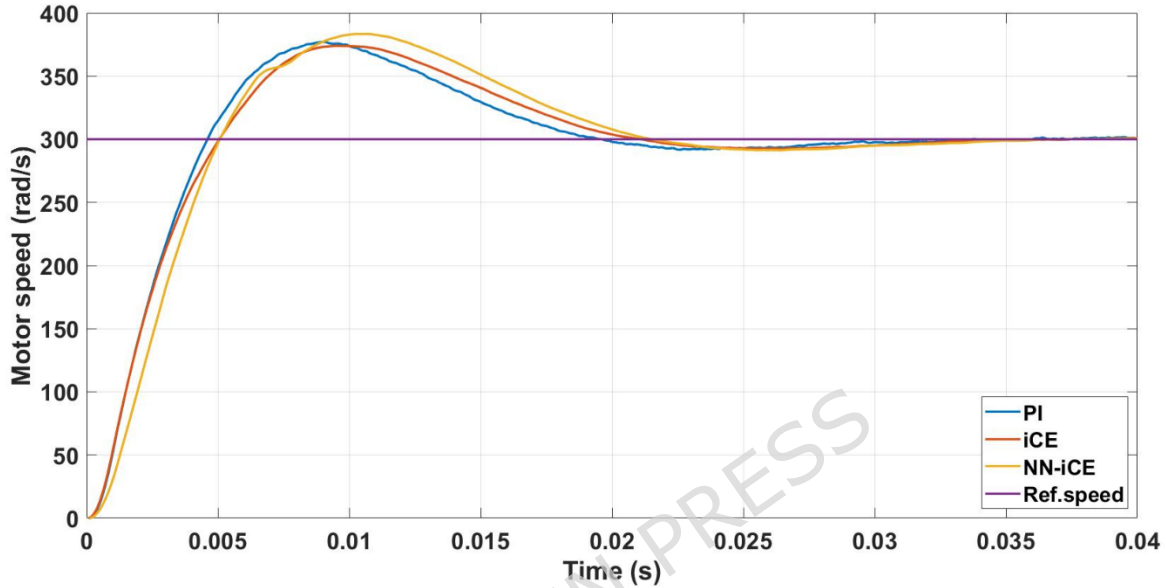
**FIGURE 14. Speed response for change in mechanical parameters ( $J_{eq}$ ,  $B_a$ )**

Fig. 14 gives the speed responses of all the models, when the mechanical parameters  $J_{eq}$ ,  $B_a$  are increased to  $24 \times 10^{-4} \text{ kg-m}^2$  and  $2 \times 10^{-4} \text{ Nm-s}$  respectively. Both the conventional model and proposed models settle slower, while transient peaks are higher due to increase in the mechanical parameters. Among the 3 models, the transient peak is least (1.4%) for the proposed iCE model, while the conventional model settles faster (0.1sec).

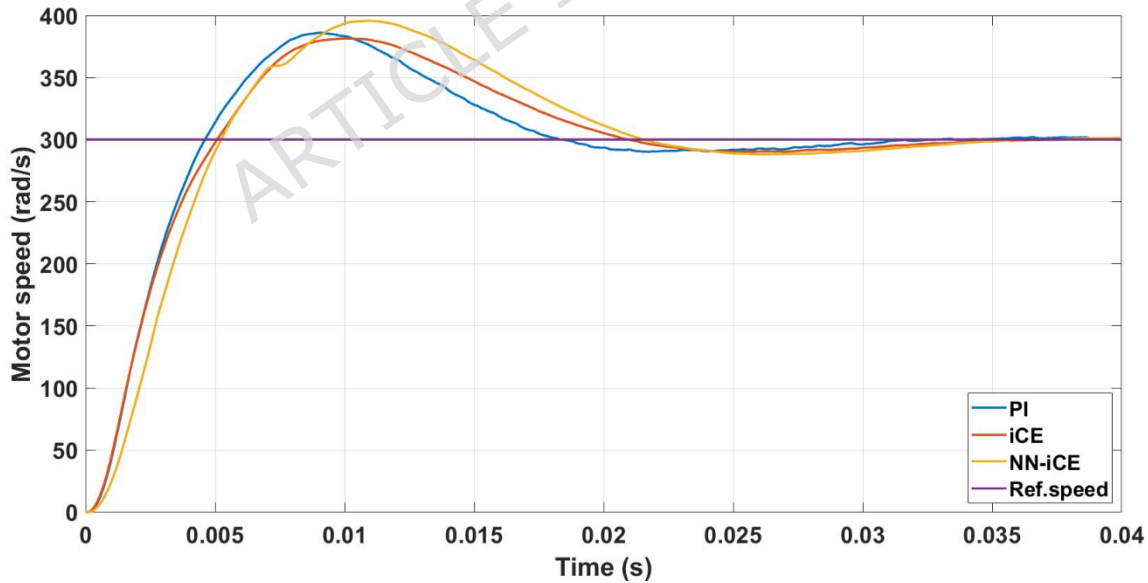
Fig. 15 gives the responses of the two proposed models, when the electrical parameters  $R$ ,  $L_{dq}$  are decreased to  $1.71 \Omega$  and  $0.0051 \text{ H}$  respectively. While the transient peaks are reduced in this case, the settling time is slightly slower compared to the rated conditions. The transient peak is least (1.247%) for the

proposed iCE model, while the conventional model settles faster (0.035sec) when the 3 models are compared.

Fig. 16 gives the responses of the two proposed models, when the rotor magnetic flux linkage is increased to 0.2167 Wb-turns. The iCE model has the least transient peak (1.271%) while conventional model has the fastest settling time (0.032sec).



**FIGURE 15. Speed response for change in electrical parameters ( $R$ ,  $L_{dq}$ )**

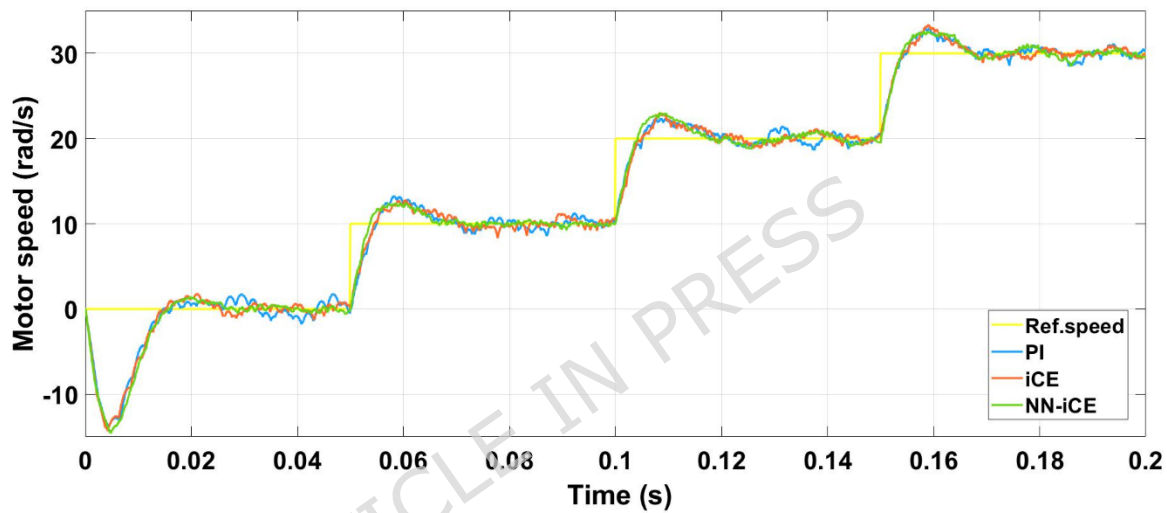


**FIGURE 16. Speed response for change in flux linkage ( $\Psi_d$ )**

TABLE 3  
COMPARISON OF SPEED RESPONSES FOR PARAMETER VARIATIONS

| VARIATIONS | DESCRIPTION | PI | iCE | NN-iCE |
|------------|-------------|----|-----|--------|
|------------|-------------|----|-----|--------|

|                                 |                    |              |              |       |
|---------------------------------|--------------------|--------------|--------------|-------|
| Mechanical parameters (Fig. 14) | Peak overshoot (%) | 1.547        | <b>1.4</b>   | 1.497 |
|                                 | Settling time (s)  | <b>0.1</b>   | 0.105        | 0.115 |
| Electrical parameters (Fig. 15) | Peak overshoot (%) | 1.257        | <b>1.247</b> | 1.278 |
|                                 | Settling time (s)  | <b>0.035</b> | 0.037        | 0.037 |
| Magnetic flux (Fig. 16)         | Peak overshoot (%) | 1.287        | <b>1.271</b> | 1.319 |
|                                 | Settling time (s)  | <b>0.032</b> | 0.036        | 0.036 |

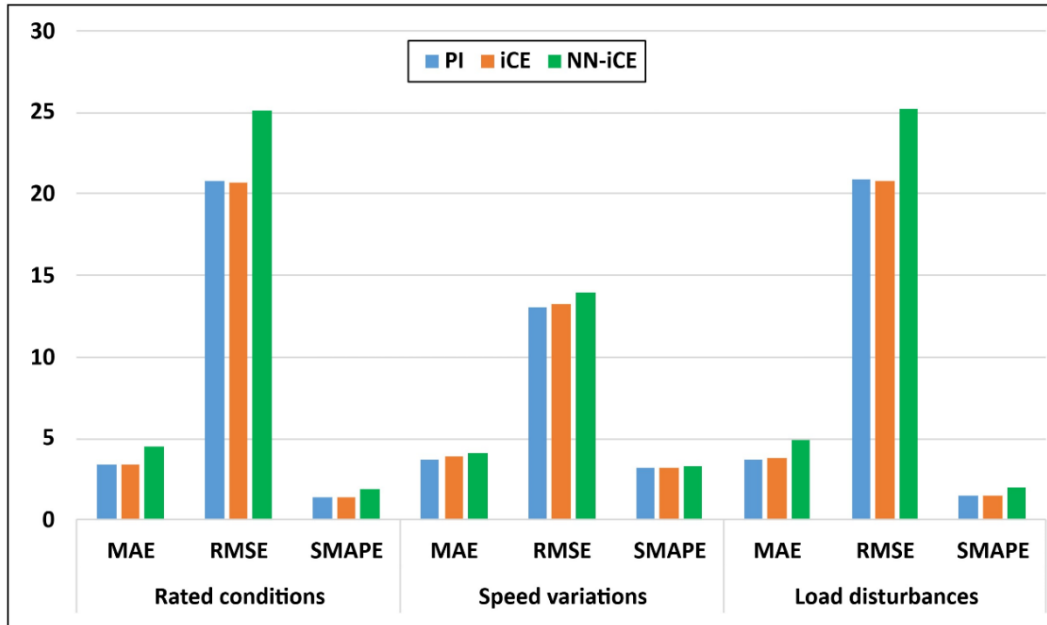


**FIGURE 17. Speed response at zero-speed and low-speed operation**

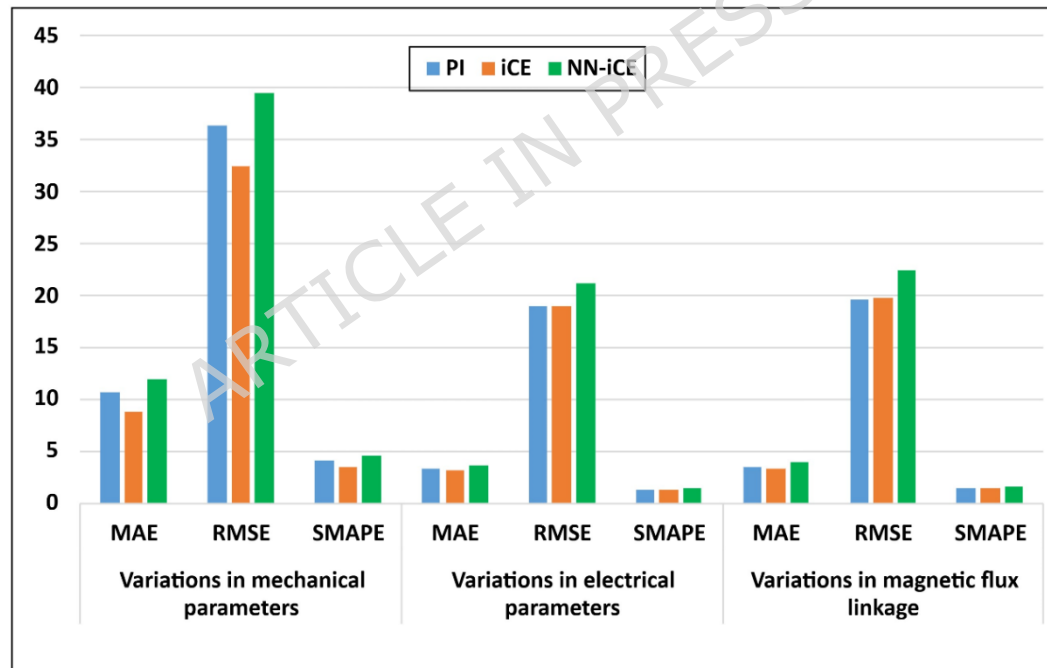
##### 5) LOW-SPEED AND ZERO-SPEED OPERATIONS

The speed responses of the proposed models for zero-speed and low-speed operations is given by Fig. 17. When the motor is started at zero speed, there is a speed dip to -14 rad/s, which can be due to sudden introduction of torque, initial rotor angle error, etc. As the drive model used in this study does not include high frequency signal injection or zero-speed observer, this negative speed is unavoidable. At 0.015sec, the conventional model and the proposed models attain the desired speed, i.e. zero. The proposed models settle faster and smoother compared to the conventional model. This shows that the proposed models reject the initial transient disturbance effectively, the current loops are reliable, and improve the zero-speed behavior.

Between zero speed and low speed levels (0-10% of rated speed), the proposed iCE and NN-iCE models have better damping and reduced oscillations compared to the conventional model.



**FIGURE 18. Evaluation metrics - Table 4 (fixed parameters)**



**FIGURE 19. Evaluation metrics - Table 4 (parameter variations)**

TABLE 4

EVALUATION METRICS FOR SPEED RESPONSE

| Metri<br>cs         |           | PI      | iCE            | NN-iCE  |
|---------------------|-----------|---------|----------------|---------|
| Rated<br>conditions | MAE       | 3.4225  | <b>3.3904</b>  | 4.4981  |
|                     | RMSE      | 20.8228 | <b>20.6936</b> | 25.1260 |
|                     | SMAP<br>E | 1.4118  | <b>1.4036</b>  | 1.8706  |

|                                     |        |                |                |         |
|-------------------------------------|--------|----------------|----------------|---------|
| Speed variations                    | MAE    | <b>3.7325</b>  | 3.8669         | 4.1151  |
|                                     | RMSE   | <b>13.0216</b> | 13.2835        | 13.9465 |
|                                     | SMAP E | 3.1963         | <b>3.1616</b>  | 3.3489  |
| Load disturbances                   | MAE    | <b>3.7288</b>  | 3.7692         | 4.8839  |
|                                     | RMSE   | 20.8869        | <b>20.7627</b> | 25.1854 |
|                                     | SMAP E | <b>1.5131</b>  | 1.5289         | 1.9982  |
| Variations in mechanical parameters | MAE    | 10.6983        | <b>8.7954</b>  | 11.8519 |
|                                     | RMSE   | 36.3095        | <b>32.3718</b> | 39.4251 |
|                                     | SMAP E | 4.0042         | <b>3.3997</b>  | 4.5732  |
| Variations in electrical parameters | MAE    | 3.2425         | <b>3.0906</b>  | 3.5634  |
|                                     | RMSE   | <b>18.8818</b> | 18.9855        | 21.1965 |
|                                     | SMAP E | 1.3153         | <b>1.2628</b>  | 1.4770  |
| Variations in magnetic flux linkage | MAE    | 3.4373         | <b>3.3500</b>  | 3.9563  |
|                                     | RMSE   | <b>19.5250</b> | 19.7441        | 22.4050 |
|                                     | SMAP E | 1.3878         | <b>1.3543</b>  | 1.6174  |

### B. EVALUATION METRICS

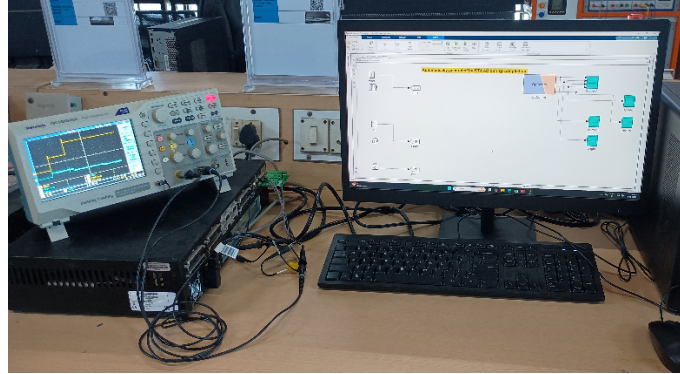
The performance of the proposed models is evaluated against the conventional model with the help of evaluation or performance metrics such as MAE, RMSE and SMAPE. Table 4, Fig. 18 and Fig. 19 give the metrics for the speed responses in the previous subsection. The least errors in each set is highlighted in bold. As seen in the table, the proposed iCE model gives least errors in majority of the metrics (67%), compared to the conventional model.

### C. REAL-TIME RESULTS

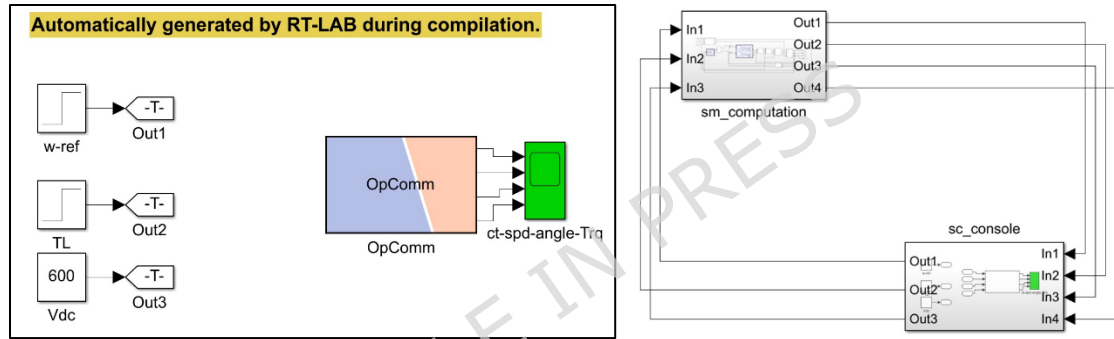
The proposed models are tested in real-time environment using the Opal-RT device OP4512 [27-28]. Fig. 20 gives the Opal-RT setup on Software-in-loop (SIL) mode with OP4512 device and digital oscilloscope. The simulation model that was used to test the models is shown in Fig. 21.

Fig. 22 to Fig. 24 show the SIL results of the models, for the conditions in subsection B. The starting speed is set at 150 rad/s and increased to 300 rad/s at 0.5 sec timestep. The motor is started on full load and reduced to half load at 0.55 sec timestep. When speed increases at 0.5 sec, it is reflected in the torque and speed responses of both the proposed models. When load decreases at 0.55 sec, for both the proposed models, sudden increase in speed

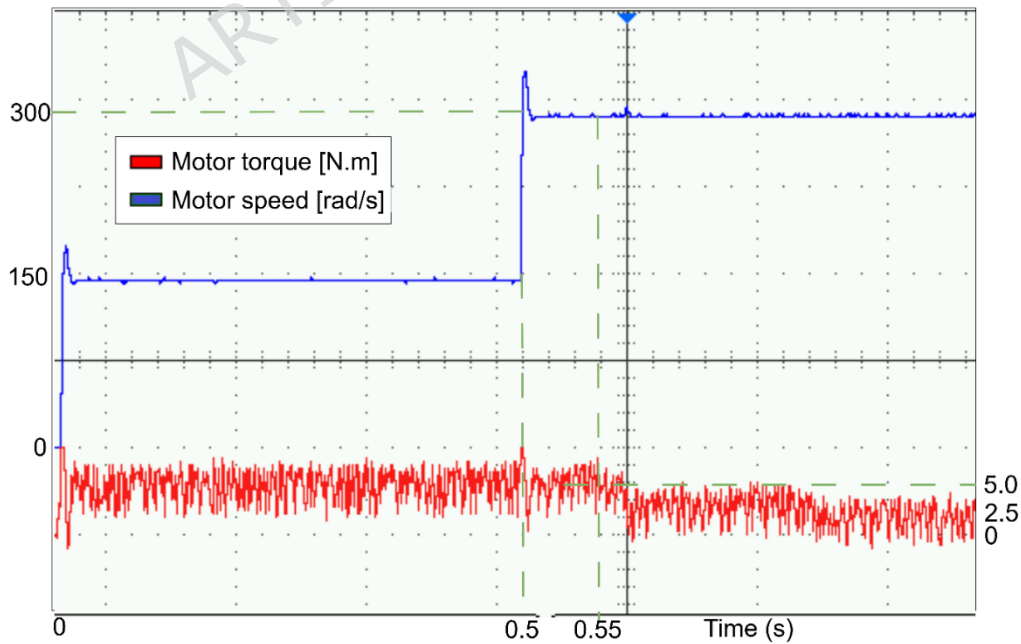
response is visible along with change in torque response. Fig. 25 gives the 3-phase current response of the proposed model for the changes in speed and load.



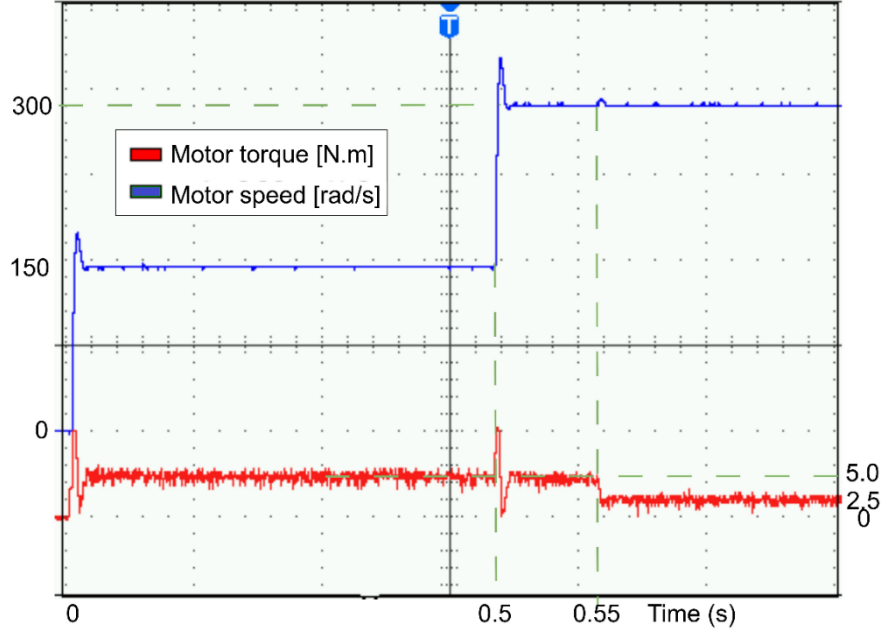
**FIGURE 20. Opal-RT setup for SIL**



**FIGURE 21. Simulation model in RT-LAB platform**



**FIGURE 22. Conventional model: Speed and torque responses**



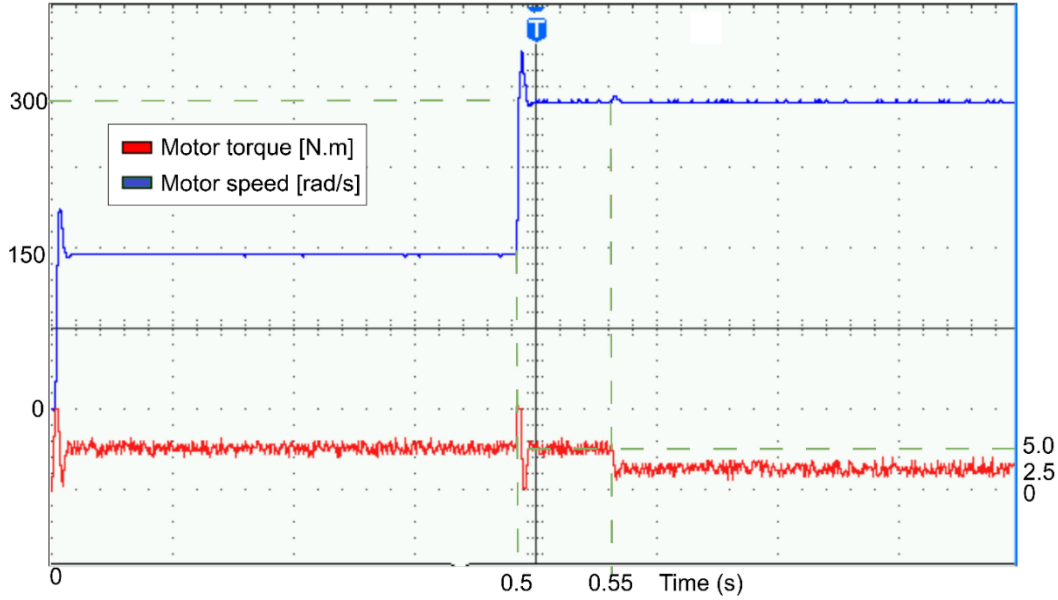
**FIGURE 23. iCE model: Speed and torque responses**

## V. RESULT DISCUSSION

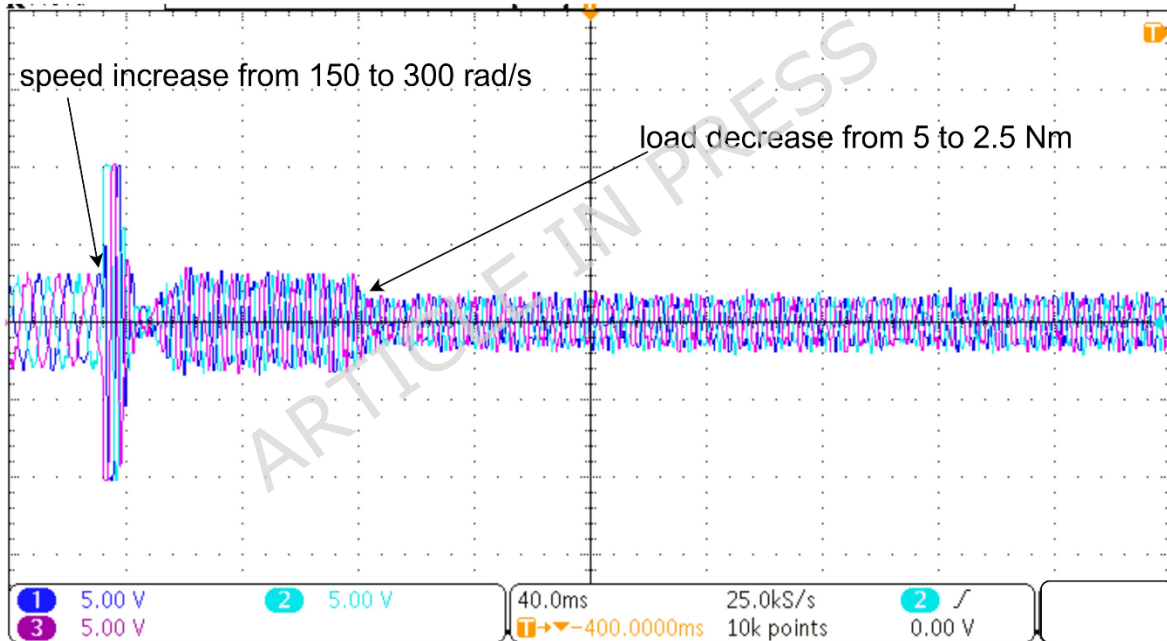
MATLAB simulation and the Real-Time environment are used to validate the proposed models under various test scenarios. The evaluation takes into account the drive's estimated rotor angle, speed response, and current and torque responses.

### *A. SIMULATION RESULTS*

According to the assessment of the total drive response under rated conditions, load disturbances, and speed fluctuations, the responses of the proposed iCE models that use ab-axis system are very similar to that of the conventional FOC model that uses dq-axis system for current control.



**FIGURE 24. NN-based iCE model: Speed and torque responses**



**FIGURE 25. Current response of proposed model**

It is evident from the response graphs (Fig. 7 – Fig. 25) as well as the performance metrics (Table 3).

As seen from Table 2 and speed responses (Figures 7, 11, 13 – 16), the proposed iCE model has lesser speed overshoot (1.275% or 382.75 rad/s) while proposed NN-based iCE model has slightly higher value (1.359% or 407.83 rad/s), compared to the conventional model (1.315% or 394.5 rad/s). The settling times of estimated speed responses vary by 4 milliseconds, which is negligible as seen in Fig. 11. A sudden increase or decrease of load results in same response time (0.015 sec) from all the models, as shown by Fig. 13. For

parameter variations, the speed responses of proposed models are similar to that of the conventional model. Comparing all the results in this case, responses of the proposed iCE model lag the conventional model responses by only 2-5 milliseconds, and gives least transient peaks in all the variation tests.

For torque responses, as seen from Fig. 8 and 12, transient peaks are lesser for the proposed models. Though a settling delay of 0.018 sec occurs compared to conventional model, once the responses have settled, all the models perform alike for every disturbance. Fig. 13 shows a better current response for proposed models when the PI controllers in the current and speed loops of the conventional and proposed models are set with same values. While the models have the same settling time for the currents, proposed models draw lesser current for the same specifications, with lesser transient peaks when compared to the conventional model.

In practical cases, the measurement noise, current distortion, and inverter non-idealities can all have an impact on the rotor angle's inverse trigonometric computation. Because the arctangent function is nonlinear, slight errors in observed  $\alpha\beta$  currents can produce significant changes in the calculated angle, especially during transient operating situations where currents fluctuate quickly. Current distortion induced by harmonics, inverter dead time, and switching effects can bring additional bias or ripple into the angle estimation. These effects can be mitigated by using proper current filtering, accurate current sensing, and synchronized sampling, which are typical in motor drive systems. As seen in Fig. 10, the estimated rotor angles of proposed models lag initially during transient state, due to current distortions. But they match the conventional model values after a slight time delay (0.02 sec).

One main limitation of the proposed iCE model is the range of PI values, similar to the conventional model. This is where the proposed ML-based iCE model gains significance. It reduces the burden of the PI tuning in FOC model. Machine learning does not require the exact mathematical model of the system; it uses domain knowledge to design the ML models. However, it requires training and testing data which need to be collected properly, as the ML modelling depends on it. Finding a good-fit model can be time-consuming depending on the algorithm or network selected and features of the available ML designing system. This can be overcome with the help of graphical processing units.

The NN model proposed in this paper is an initial work of implementing ML-based iCE model. Neural networks have the capability to extract complex patterns from given data and hence, a NN-based FOC model enhances robustness and control precision, and is adaptable even under non-ideal conditions. However, the computational burden of neural networks is an

obstacle for FOC which run on short sampling times (or high switching frequencies), and can affect the performance of NN-based FOC. The multiple layers of computation in the NN model require extensive mathematical calculations in every control loop iteration, which significantly increases the computational load and also limits the inference time. If the inference time exceeds the sampling time frame, it can lead to degradation of model performance. This can be overcome to some extent by optimization techniques such as hyperparameter optimization, pruning, quantization etc., and by using control platforms with high-speed parallel processing such as multi-core processors, FPGAs (Field Programmable Gate Arrays) and GPUs (Graphics Processing Units).

The proposed NN-iCE model gives performance comparable with the conventional model and the proposed iCE model. However, it is less efficient, which can be attributed to the computational burden of NN. The NN-iCE model requires more finetuning, larger dataset and an upgraded control platform to improve its performance.

TABLE 5  
COMPARISON WITH EXISTING WORK

| Ref. | Paper Highlights                                      | Current<br>Controller<br>Type | Controller<br>Ref.<br>Frame<br>Type | Speed<br>Estimation<br>Type     |
|------|-------------------------------------------------------|-------------------------------|-------------------------------------|---------------------------------|
| [5]  | Sensorless speed control of PMSM                      | PI                            | Dq                                  | SMO                             |
| [6]  | Unified FOC controller for PMSM                       | PI                            | Dq                                  | -                               |
| [7]  | ML-based current controller                           | LR / SVM /<br>NN / LSTM       | Dq                                  | -                               |
| [8]  | Sensorless FOC for PMSM                               | PI                            | Dq                                  | Angle tracking<br>PLL           |
| [9]  | Sensorless FOC                                        | PI                            | Dq                                  | SMO                             |
| [10] | FOC of PMSM                                           | PI                            | Dq                                  | Tracking &<br>BEMF<br>observers |
| [11] | NN-based rotor position estimation                    | PI                            | Dq                                  | NN & PLL                        |
| [12] | Sensorless speed control with single current sensor   | PI                            | Dq                                  | MRAS                            |
| [13] | Speed and position estimator                          | PI                            | Dq                                  | Adaptive<br>mechanism           |
| [14] | Estimation of rotor position in PMSM                  | PI                            | Dq                                  | BEMF & MRAS                     |
| [17] | Vector control and hall-effect sensors                | PI                            | Dq                                  | -                               |
| [18] | PMSM Vector control                                   | PI                            | Dq                                  | -                               |
| [19] | Integrated electric drive system for electric vehicle | PI                            | dq                                  | -                               |

|                 |                                                 |                |           |                           |
|-----------------|-------------------------------------------------|----------------|-----------|---------------------------|
| [20]            | IPMSM drive control technology                  | PI             | dq        | -                         |
| [21]            | Brain Emotional Controller for speed control    | PI             | dq        | -                         |
| <b>Proposed</b> | <b>Integrated Controller Estimator for PMSM</b> | <b>PI / NN</b> | <b>ab</b> | <b>Mathematical model</b> |

### ***B. REAL-TIME RESULTS***

Studying the controllers using a real-time physical device in a SIL environment, both the proposed models perform satisfactorily during speed change and load change. For the same values of PI, the conventional model fails to reflect the load variation while the proposed models reflect it satisfactorily. The speed change is reflected by all the models.

### ***C. COMPARISON WITH EXISTING WORK***

Table 5 gives a comparison of proposed approach with the existing literature. The FOC techniques use dq-axis or rotating reference frame system for current control. Various speed estimation techniques like PLL, SMO, MRAS and BEMF observer are available with different complexity levels. Very few methods like [13] use dq-axis system for estimation purposes while majority use ab-axis or stationary reference frame system. Most of the estimation techniques require voltage and current sensors. In all the cases, the current control and estimation functions are performed as separate units.

The proposed approach integrates the current control and estimation functions as a single unit, by employing the ab-axis system for the purpose. In addition, only current sensors are required for the functions of control and estimation. The currents in ab-axes are directly proportional to trigonometric functions of rotor angle, and speed is derived from the rotor angle. This reduces the complexity of estimation method. By using the same input set for both

functions, the circuit becomes more compact. While in [12] a single current sensor is used, the purpose is only speed estimation. The proposed method performs both current control and estimation.

The proposed work is extended with machine learning by introducing the proposed NN-based model, which incorporates the rapidly expanding field of Artificial Intelligence.

## **VI. CONCLUSION**

This paper presents a novel method for sensorless field-oriented control of PMSM drives. The main objective is to improve the existing sensorless FOC techniques by integrating the speed control and estimation functions. The secondary objective is to explore the possibility of ML application in the proposed technique. The performances of the proposed models are compared

with a conventional model under various test conditions and validated using simulation and in real-time.

In terms of different drive responses, the transient peaks are lesser for the proposed iCE model (reduced by 0.04% for speed, 0.09% for torque) and requires lesser current for the same drive specifications. The response time for speed variations (0.025 sec) and torque variations (0.015 sec) are same as that of the conventional model. In terms of evaluation metrics, the proposed model performs better for more metrics (67%) than the conventional model. The proposed NN-based model give performance comparable with the other models, but less efficient. Further improvement of the performance of the proposed NN-based iCE model can be achieved by adjusting the training parameters and utilizing a larger data set.

### **Data Availability**

The corresponding author can provide the datasets used and/or analyzed in the current work upon reasonable request.

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### **REFERENCES**

- [1] S. J. Rind, Y. Ren, et.al., "Configurations and control of traction motors for electric vehicles: A review," in Chinese Journal of Electrical Engineering, vol. 3, no. 3, pp. 1-17, December 2017.
- [2] C. Liu and Y. Luo, "Overview of advanced control strategies for electric machines," in Chinese Journal of Electrical Engineering, vol. 3, no. 2, pp. 53-61, September 2017, doi: 10.23919/CJEE.2017.8048412.
- [3] Özçiflikçi, O.E., Koç, M., Bahçeci, S. et al. Overview of PMSM control strategies in electric vehicles: a review. Int. J. Dynam. Control 12, 2093-2107 (2024).
- [4] M. Monadi, M. Nabipour, F. Akbari-Behbahani and E. Pouresmaeil, "Speed Control Techniques for Permanent Magnet Synchronous Motors in Electric Vehicle Applications Toward Sustainable Energy Mobility: A Review," in IEEE Access, vol. 12, pp. 119615-119632, 2024.
- [5] Kung, YS., Risfendra, R., Lin, YD. et al., "FPGA-realization of a sensorless speed controller for PMSM drives using novel sliding mode observer, " in Microsystem Technologies, vol. 24, pp. 79-93, 2018, doi: 10.1007/s00542-016-3179-6
- [6] M. Koç, "Unified Field Oriented Controlled Drive System for All Types of PMSMs Considering System Nonlinearities," in IEEE Access, vol. 10, pp. 56773-56784, 2022, doi: 10.1109/ACCESS.2022.3178104.

- [7]Tom, A. M., & Daya, J. L. (2025). Design of machine learning-based controllers for speed control of PMSM drive. *Scientific Reports*, 15(1), 1-24.
- [8]Prasad Kulkarni, "Sensorless Field-Oriented Control of Permanent Magnet Synchronous Motor (Surface and Interior) for Appliances with Angle-Tracking Phase-Locked Loop Estimator," Microchip Technology Inc., 2019 [Online] Available: <https://ww1.microchip.com/downloads/en/DeviceDoc/TB3220-Sensorless-Field-Oriented-Control-of-PMSM-for-Appliances-DS90003220A.pdf> Accessed on : July 16,2024.
- [9]Infineon Technologies AG., "Sensorless field-oriented control (FOC) usingPSoC™ 6 MCU," 2022 [Online]. [https://www.infineon.com/dgdl/Infineon-AN235096\\_Sensorless\\_field-oriented\\_control\\_FOC\\_using\\_PSoC\\_6\\_MCU-ApplicationNotes-v01\\_00-EN.pdf?fileId=8ac78c8c821f280601821f2bea270000](https://www.infineon.com/dgdl/Infineon-AN235096_Sensorless_field-oriented_control_FOC_using_PSoC_6_MCU-ApplicationNotes-v01_00-EN.pdf?fileId=8ac78c8c821f280601821f2bea270000) Accessed on: July 11,2024.
- [10] NXP Semiconductors, "MCUXpresso SDK Field-Oriented Control (FOC) of 3-Phase PMSM and BLDC Motors - Userguide," 2023 [Online]. Available: <https://www.nxp.com/docs/en/user-guide/PMSMMCXN9XXEVK.pdf> Accessed on: July 16,2024.
- [11] W. Zine, Z. Makni, E. Monmasson, L. Idkhajine and B. Condamin, "Interests and Limits of Machine Learning-Based Neural Networks for Rotor Position Estimation in EV Traction Drives," in *IEEE Transactions on Industrial Informatics*, vol. 14, no. 5, pp. 1942-1951, May 2018, doi: 10.1109/TII.2017.2765398.
- [12] S. S. Badini, V. Verma, M. Tariq and S. Urooj, "Single Current Sensor-Based Speed Sensorless Vector Controlled PMSM Drive," in *IEEE Access*, vol. 11, pp. 86362-86376, 2023. doi: 10.1109/ACCESS.2023.3266425.
- [13] J. Mat Lazi, Z. Ibrahim, M.H.N. Talib, A. Alias, A. Hanafi and M. Azri, "Speed and Position Estimator of for Sensorless PMSM Drives using Adaptive Controller," *International Journal of Power Electronics and Drive Systems (IJPEDS)*, vol. 10, no. 1, pp. 128-136, 2019. doi: 10.11591/ijpeds.v10.i1.pp128-136.
- [14] Zhang, Hua Qiang, et al. "An Improved Estimation Method of Permanent Magnet Synchronous Motor without Position Sensor." *Advanced Materials Research*, vol. 159, Trans Tech Publications, Ltd., pp. 616-621 Dec. 2010. doi:10.4028/www.scientific.net/amr.159.616.

- [15] I.M. Abdelaziz, Y.S. Mohamed, E.G. Shehata, et al. "Comprehensive analysis of IPMSM drives with FOC under different operating modes." *Discover Applied Sciences* vol. 7, no. 4, pp. 1-38, 2025.
- [16] G. Wang, M. Valla, and J. Solsona, "Position sensorless permanent magnet synchronous machine drives—A review," in *IEEE Transactions on Industrial Electronics*, vol. 67, no. 7, pp. 5830–5842, Jul. 2020.
- [17] R. R. Sorial, M. H. Soliman, H. M. Hasanien and H. E. A. Talaat, "A Vector Controlled Drive System for Electrically Power Assisted Steering Using Hall-Effect Sensors," in *IEEE Access*, vol. 9, pp. 116485-116499, 2021.
- [18] M. Brejl and M. Princ, "Using the PMSM Vector Control," Freescale Semiconductor, 2012 [Online]. Available: <https://www.nxp.com/docs/en/application-note/AN2972.pdf> Accessed on: Jun.5, 2022.
- [19] J. Hu, T. Peng, M. Jia, Y. Yang and Y. Guan, "Study on Electromechanical Coupling Characteristics of an Integrated Electric Drive System for Electric Vehicle," in *IEEE Access*, vol. 7, pp. 166493-166508, 2019.
- [20] L. Li and Q. Liu, "Research on IPMSM Drive System Control Technology for Electric Vehicle Energy Consumption," in *IEEE Access*, vol. 7, pp. 186201-186210, 2019.
- [21] MD. Qutubuddin, and N. Yadaiah, "Modeling and implementation of brain emotional controller for Permanent Magnet Synchronous motor drive," *Engineering Applications of Artificial Intelligence*, Vol. 60, pp. 193-203, 2017.
- [22] J. L. F. Daya & A. M. Tom, "System for Regulating Speed of Electric Motor based on Current Feedback Signals, and Method thereof", Patent Application Publication India 202441050015A, June 29, 2024.
- [23] S. Shalev-Shwartz and S. Ben-David, *Understanding Machine Learning: From Theory to Algorithms*. Cambridge: Cambridge University Press, 2014.
- [24] F. Jaffar, T. Farid, M. Sajid, Y. Ayaz and M. J. Khan, "Prediction of Drag Force on Vehicles in a Platoon Configuration Using Machine Learning," in *IEEE Access*, vol. 8, pp. 201823-201834, 2020.
- [25] J. Duan and H. Kashima, "Learning to Rank for Multi-Step Ahead Time-Series Forecasting," in *IEEE Access*, vol. 9, pp. 49372-49386, 2021.
- [26] P. Asthana, S. Mishra, N. Gupta, M. Derawi and A. Kumar, "Prediction of Student's Performance With Learning Coefficients Using Regression Based Machine Learning Models," in *IEEE Access*, vol. 11, pp. 72732-72742, 2023.

- [27] M. X. Bui, R. Dutta and F. Rahman, "Application of Deep Learning in Parameter Estimation of Permanent Magnet Synchronous Machines," in IEEE Access, vol. 12, pp. 40710-40721, 2024, doi: 10.1109/ACCESS.2024.3377224.
- [28] J. S. Lee and G. Choi, "Modeling and hardware-in-the-loop system realization of electric machine drives — A review," in CES Transactions on Electrical Machines and Systems, vol. 5, no. 3, pp. 194-201, Sept. 2021, doi: 10.30941/CESTEMS.2021.00023.

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